

Architecture overview

A technical map of how Priceplain turns founder assumptions into pricing, metering, revenue simulation, Solvimon handoff, Codeplain specs and sovereign AI review.

Purpose. This PDF gives judges and collaborators a single technical overview of Priceplain: the browser app, deterministic engines, serverless AI routes, sponsor integrations, data flows and security boundaries.

MVP scope. Priceplain is a planning and handoff layer. It does not process payments directly and does not claim a production Solvimon integration. It creates the pricing, metering and governance artefacts needed before billing implementation.

Solvimon

Usage-based pricing, hybrid plans, meters, invoice items, credits and checkout handoff.

Codeplain

.plain files define product scope, pricing rules and acceptance-test behaviour.

Vercel

Deployed front end plus serverless API routes for AI/provider calls.

FLock

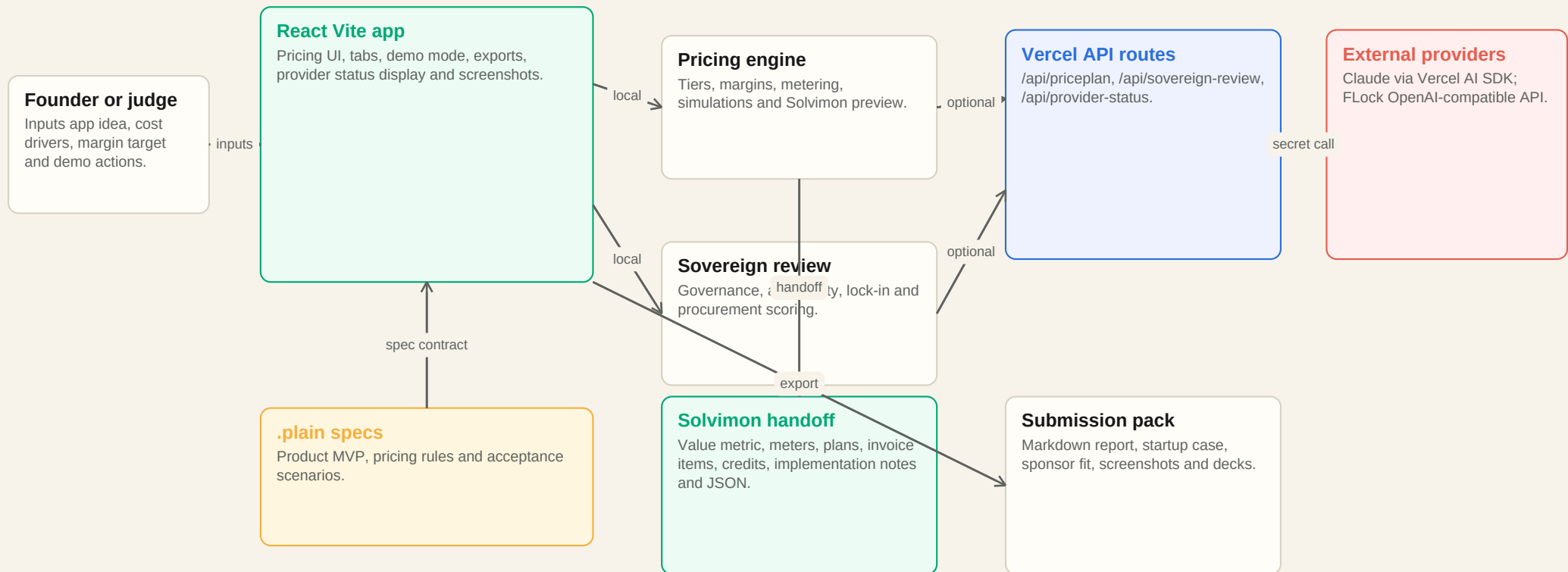
Sovereign AI review path for data ownership, auditability, lock-in and procurement risk.

Key takeaways

- Most business logic runs deterministically in the browser, so the demo works without provider credentials.
- Provider keys stay server-side in Vercel API routes and are never exposed to the browser.
- Solvimon output is an implementation-ready handoff: meters, plans, invoice items, credits and JSON.
- Codeplain evidence lives in plain/product-mvp.plain, plain/pricing-engine.plain and plain/acceptance-tests.plain.

System context and component architecture

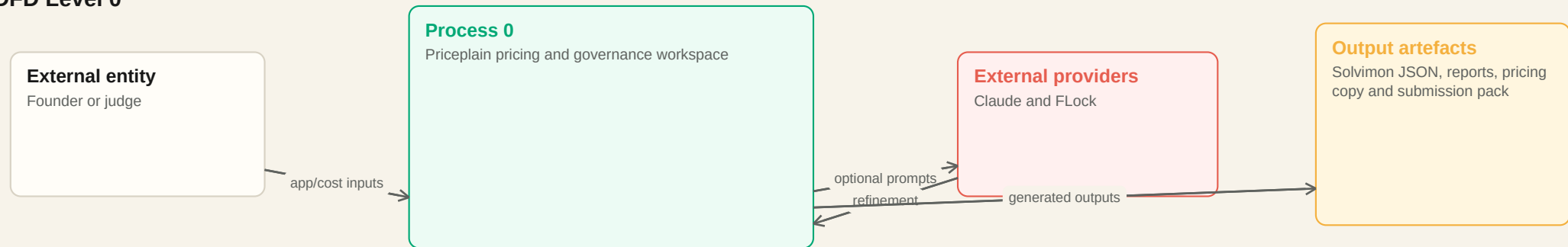
The browser owns the interactive workspace and deterministic engines. Serverless routes are only used for optional AI refinement, provider readiness and secret-protected calls.



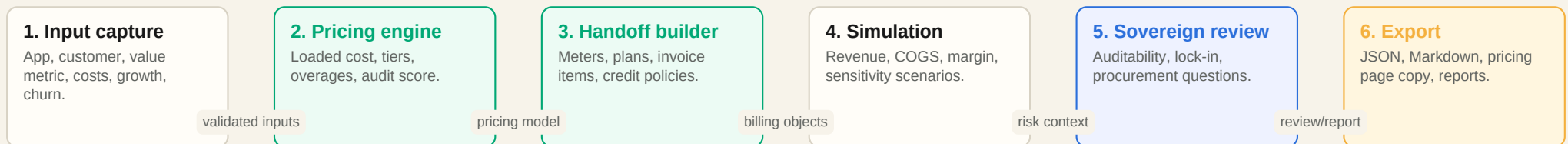
Data Flow Diagram

DFD view of the core data movement. Sensitive keys are environment variables on serverless routes, not browser-visible data.

DFD Level 0

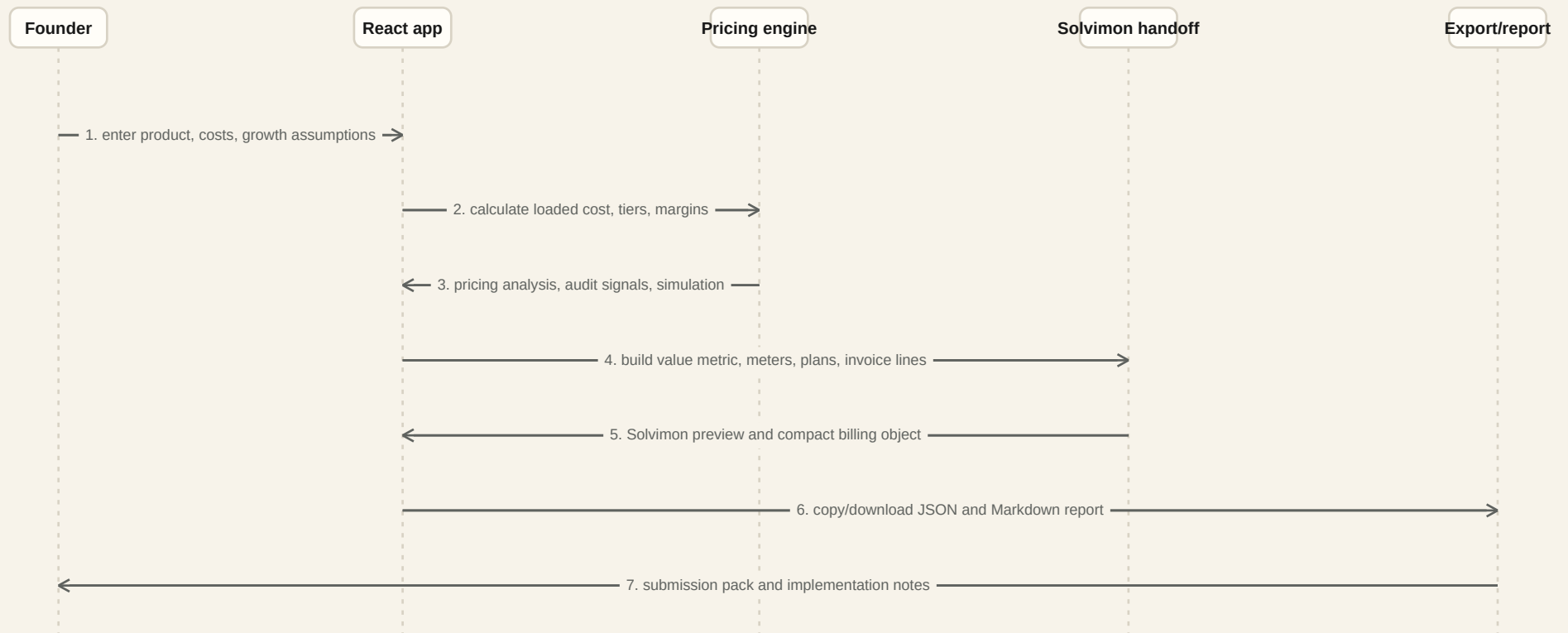


DFD Level 1



Sequence flow: pricing to Solvimon handoff

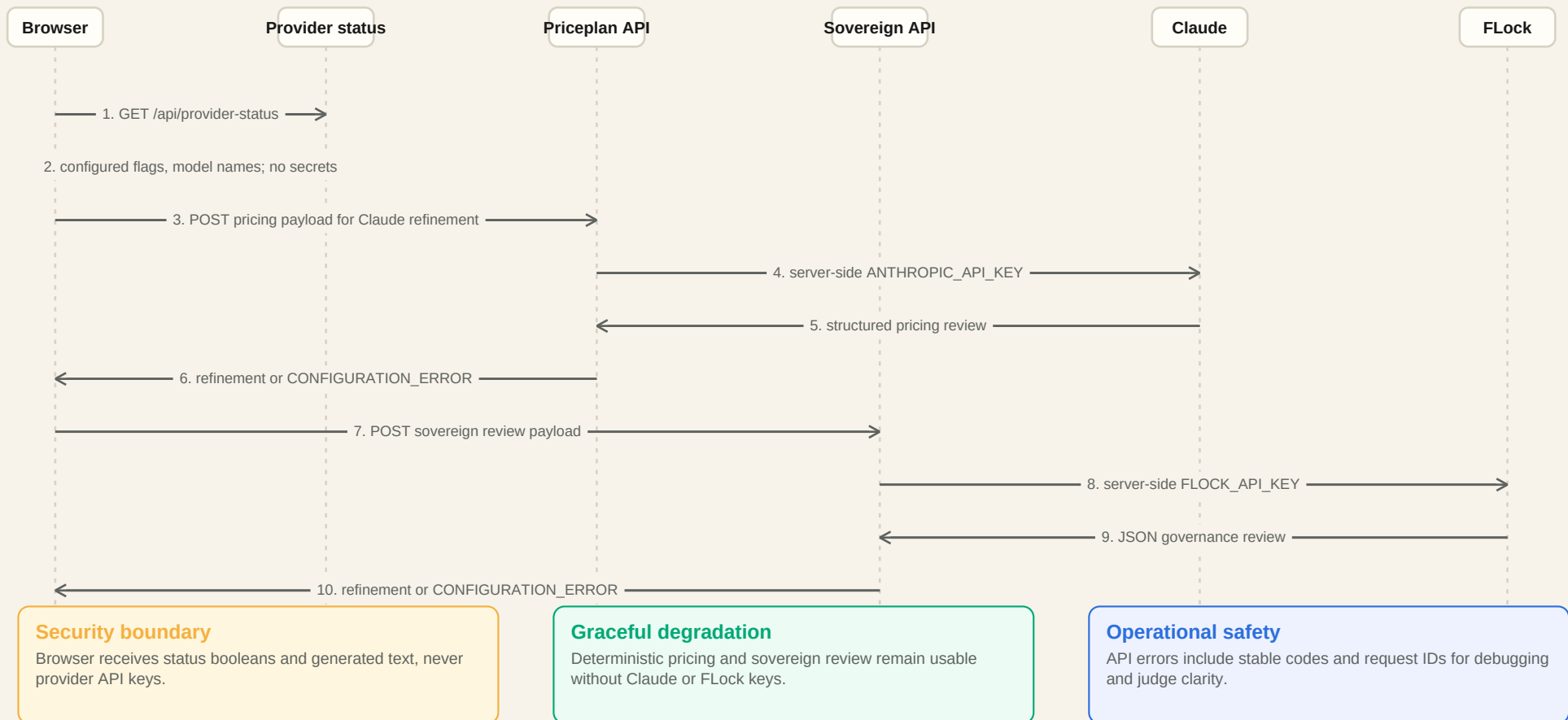
This is the main judge demo path. The deterministic engine works offline; Solvimon output is a ready-to-implement handoff rather than live payment processing.



Design decision. Priceplain does not collect payments. It creates the pricing and billing object plan that can be implemented in Solvimon: value metric, meter events, plans, invoice items, credits, overages and checkout guidance.

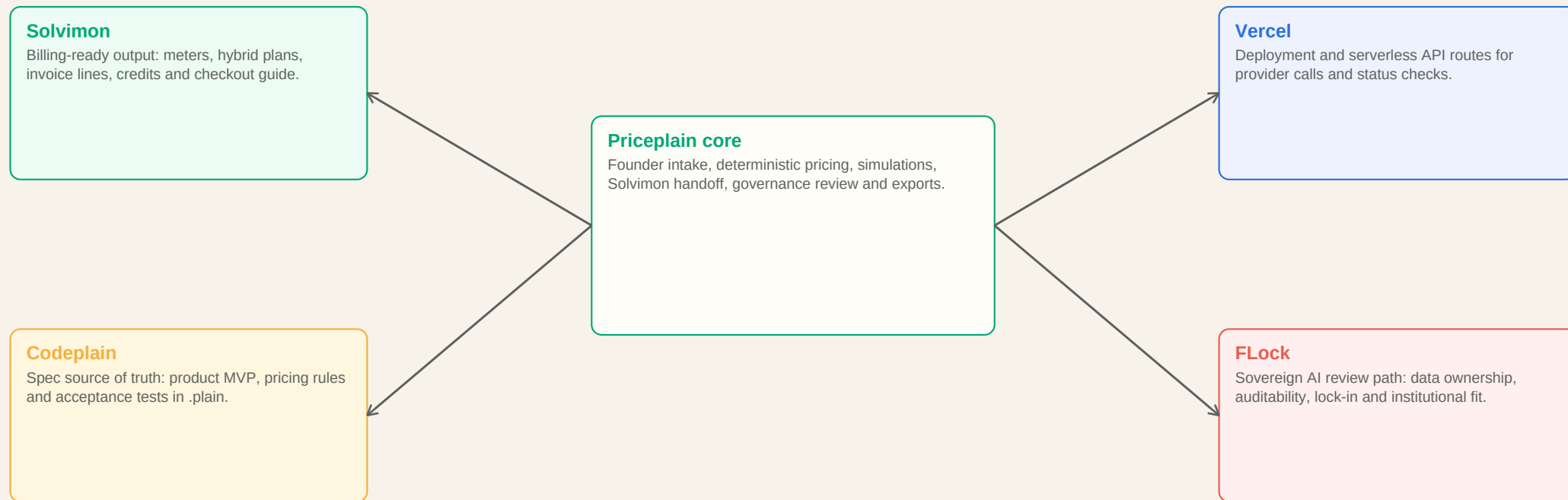
Sequence flow: optional AI providers and security boundaries

Claude and FLock are optional refinements. The core demo degrades gracefully when keys are missing, and provider secrets remain server-side.



Sponsor responsibility map

Each sponsor role is explicit. Solvimon and Codeplain are primary; Vercel and FLock are secondary technical fits.



Primary story. Codeplain keeps the AI build spec-driven. Priceplain uses those specs to generate practical pricing and metering logic. Solvimon is the natural downstream system for turning that plan into billing, checkout and revenue operations.

Secondary story. Vercel hosts the app and API routes. FLock supports the sovereign AI governance angle by giving teams a privacy-conscious route for institutional review and provider-risk thinking.

Operational views for judges and maintainers

Useful artefacts, trust controls and extension points for taking the MVP beyond the hackathon.

Runtime paths

- / - pricing workspace
- /?tab=metering - Solvimon handoff
- /?tab=simulation - revenue simulation
- /?tab=business - startup case
- /?tab=sovereign - FLock-aligned review
- /?tab=export - submission pack

Trust controls

- Deterministic baseline works offline
- Provider keys stay server-side
- Provider status exposes booleans only
- Stable error codes and request IDs
- Exported JSON/Markdown for review
- No direct payment processing in MVP

Next architecture steps

- Create live Solvimon sandbox adapter
- Persist saved workspaces
- Add pricing decision audit history
- Add organisation profiles for FLock review
- Add benchmark preset library
- Add team collaboration and report sharing

Submission artefacts

docs/SOLVIMON_STARTUP_PLAN.md, docs/FLOCK_SOVEREIGN_AI_ONE_PAGER.md, docs/CODEPLAIN_ONE_PAGER.md, docs/Priceplain_Canva_Deck.pptx, plain/product-mvp.plain, plain/pricing-engine.plain, plain/acceptance-tests.plain.